ECE 3340 Numerical Methods

Homework 1: Prerequisite Mathematics

Name: **ID:**

Solve the following problems from **Chapter 1, Prerequisites: Mathematics**. Use any available space to work out the problem and **place your final solution in the box provided**.

Problem 1: Calculate $y = z^2 z^*$ where $z = (2 + 3i)$

$y = 26 + 39i$

conjugate

$y = (2+3i)^2 \cdot (2+3i)^*$

$(4 + 12i + 9i^2)(2 - 3i)$

$i^2 = -1$

$(4 + 12i - 9)(2 - 3i)$

$(-5 + 12i)(2 - 3i)$

$-10 + 24i + 15i + 36$

$y = 26 + 39i$

Problem 2: Describe the Hilbert space of $\mathbf{y} = \mathbf{ABx}$ if $\mathbf{A} \in \mathbb{R}^{5\times 4}$ and $\mathbf{x} \in \mathbb{C}^7$.

Complex number is matrix vector

$A = \mathbb{R}^{5\times 4}$ matrix

$x = \mathbb{C}^{7\times 1}$ vector $\rightarrow$ $\underbrace{5\times 4 \cdot 7}_{}\times 1$

B would have to be the inner for this to function

What is the Hilbert space of **B**?

$B = \mathbb{C}^7$

$B = ?$ $\rightarrow$

Problem 3: Calculate $\mathbf{x} \cdot \mathbf{y}$

The inner product $\mathbf{x} \cdot \mathbf{y}$ for complex vectors is defined as $\mathbf{x}^T \bar{\mathbf{y}}$ (where $\bar{\mathbf{y}}$ is the complex conjugate of $\mathbf{y}$). Calculate $\mathbf{x} \cdot \mathbf{y}$ for the complex vectors:

$$\mathbf{x} = \begin{bmatrix} 2i \\ -7 + 2i \\ 3 - i \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} -1 + 2i \\ 1 - 3i \\ i \end{bmatrix}$$

① compute for $\bar{y}$

$\bar{y}_1 = -1 - 2i$

$\bar{y}_2 = 1 + 3i$

$\bar{y}_3 = -i$

② $x^T \cdot \bar{y} \rightarrow$ $2i(-1-2i) = 2i - 4i^2 = -2i + 4$

$(-7+2i)(1+3i) = -7 - 21i + 2i + 6i^2 = -13 - 19i$

$(3-i)(-i) = -3i + i^2 \rightarrow -3i - 1$

③ $(-2i+4) + (-13-19i) + (-3i-1)$

$4 - 13 - 1 = -10$

$-2i - 19i - 3i = 24i$

$x \cdot y = -24i - 10$

$x \cdot y = -10 - 24i$

Problem 4: Solve the following linear system using Gaussian Elimination

$x - 2y + 3z = 7$

$2x + y + z = 4$

$-3x + 2y - 2z = -10$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 2 & 1 & 1 & 4 \\ -3 & 2 & -2 & -10 \end{array}\right] \quad R_3 + 3R_1 \quad \begin{array}{l} -3+3=0 \\ 2-6=-4 \\ -2+9=7 \\ -10+21=11 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 2 & 1 & 1 & 4 \\ 0 & -4 & 7 & 11 \end{array}\right] \quad R_2 - 2R_1 \quad \left.\begin{array}{l} 2-2 \\ 1+4 \\ 1-6 \\ 4-14 \end{array}\right| \begin{array}{l} 0 \\ 5 \\ -5 \\ -10 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & 5 & -5 & -10 \\ 0 & -4 & 7 & 11 \end{array}\right] \frac{R_2}{5} \to \left[\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & 1 & -1 & -2 \\ 0 & -4 & 7 & 11 \end{array}\right] \quad R_3 + 4R_2 \quad \left.\begin{array}{l} -4+4 \\ 7-4 \\ 11-8 \end{array}\right| \begin{array}{l} 0 \\ 3 \\ 3 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 3 & 3 \end{array}\right] \frac{R_3}{3}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$z = 1$

$y - z = -2 \to y = -1$

$x - 2(-1) + 3(1) = 7$

$x + 5 = 7 \to x = 2$

$\boxed{\text{x} = 2 \qquad \text{y} = -1 \qquad \text{z} = 1}$

Problem 5: Calculate a 4th order Maclaurin series for $f(x) = e^x \cos x^2$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

understand

$0! = 1$ $\quad 2! = 2$

$1! = 1$ $\quad 3! = 6$

$4! = 24$

$$e^x = \sum \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} \to$$

$$\cos x = \sum \frac{(-1)^k}{(2k)!} x^{2k} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}$$

$$\cos x^2 = 1 - \frac{x^4}{2} + \frac{x^6}{24} - \frac{x^8}{720}$$

$e^x \cos x^2 =$

now combine the equations

$$1 + x + \frac{x^2}{2} + \frac{x^3}{6} \underbrace{\left(-\frac{x^4}{2} + \frac{x^4}{24}\right)}_{\frac{-11}{24}x^4}$$

$$\boxed{f(x) \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{11x^4}{24}}$$